SiLK in a Nutshell

Quick reference guide can be found here:
https://tools.netsa.cert.org/silk/SiLK-QuickRef-2019-12.pdf

Set up a working directory

1. Create the development environment, we will call it ATA in our Documents directory

   ```
   cd ~/Documents
   mkdir ATA
   cd ATA
   ```

2. Copy the files to work on into this directory

   ```
   cp 20230401.bgp.pmap ~/Documents/ATA
   cp honeynet.zip ~/Documents/ATA
   ```

3. Grab the Docker image from certcc/silklive

   ```
   sudo docker pull certcc/silklive
   ```

Set up an environment to evaluate the pcap

- Start a docker instance and assign your working directory as the data directory:

```
docker run --rm -it -v "/home/parallels/Documents/ATA:/data" certcc/silklive
```

- Put a copy of the silk.conf in your working directory:

```
cp /usr/local/share/silk/twoway-silk.conf /data/silk.conf
```

- Use yaf to turn a pcap into a SiLK readable format:

```
yaf --in /data/honeypot.pcap --out - \
    --applabel --max-payload=1500 --silk \
    | rwipfix2silk --silk-output=/data/honeypot.rw
```

Getting rid of the noise

- The IP address 137.184.35.111 is for the honeypot.
- The IP address range 192.168.150.0-255 is used for the VM
- When someone connects to the honeypot it brings up a VM and proxies the connection to the fresh VM.
- All authentication attempts will work
- Any commands sent are saved as are any files pulled down during the session.
- The honepot_logs/ include all activities/commands
- The files/ includes all artifacts and files saved to the honeypot
- The pcap only includes source/destination/duration/volume because it is encrypted traffic

What applications are listed in the sample?

```
# rwstats --fields=app --top --count 20\
/data/honeypot.rw
INPUT: 38240 Records for 5 Bins and 38240 Total
Records
OUTPUT: Top 20 Bins by Records
appli|   Records|  %Records|   cumul_%|
   22|     24640| 64.435146| 64.435146|
    0|     13580| 35.512552| 99.947699|
  443|        10|  0.026151| 99.973849|
   80|         6|  0.015690| 99.989540|
 3389|         4|  0.010460|100.000000|
```

A reference for field definitions can be found here: https://tools.netsa.cert.org/silk/rwcut.html

Who are the top talkers?

Total number of sIP observed

```
# rwstats --fields=sip --top --count 5 /data/honeypot.rw
INPUT: 38240 Records for 2422 Bins and 38240 Total
Records
OUTPUT: Top 5 Bins by Records
                sIP|   Records|  %Records|   cumul_%|
    137.184.35.111|     17317| 45.285042| 45.285042|
      92.118.39.92|      1701|  4.448222| 49.733264|
      92.118.39.90|      1185|  3.098849| 52.832113|
    185.225.74.213|       800|  2.092050| 54.924163|
      92.118.39.91|       629|  1.644874| 56.569038|
```

Who are the top responders?

```
# rwstats --fields=dip --top --count 5 /data/honeypot.rw
INPUT: 38240 Records for 2411 Bins and 38240 Total
Records
OUTPUT: Top 5 Bins by Records
                dIP|   Records|  %Records|   cumul_%|
     137.184.35.111|     20923| 54.714958| 54.714958|
     185.225.74.213|       803|  2.099895| 56.814854|
    167.172.241.154|       589|  1.540272| 58.355126|
     164.132.169.29|       452|  1.182008| 59.537134|
      139.59.62.104|       354|  0.925732| 60.462866|
```

What happens if we remove the Top Talker?

```
# rwfilter honeypot.rw --not-any-address=137.184.35.11\
--pass=stdout | rwstats --fields=sip --top --count 5
INPUT: 0 Records for 0 Bins
OUTPUT: Top 5 Bins by Records
            sIP|   Records|  %Records|   cumul_%|
```

By chaining the rwfilter command with the rwstats command we can take the honeypot.rw file and remove and references to the IP address 137.184.35.111. As we can see that all traffic in this sample had that address as either a source or a destination, so this command line example produces an empty result.

How can I find the top 50 IP addresses

- The flows are measure communications in one direction. You can use rwstats to get the information, but you have to capture both directions if you want the total volume to traffic.
- To do that you must take the top sip stats and the top dip stats and add them.
- To find the top talkers you want to identify the top sources that initiated communications, that means sip address. Capture the top 51 because we know 137.184.35.111 is the honeypot or filter it out with --not-saddr instead of --not-any-address.

How do I limit my output to a specific group of IP addreses

- Extract your list of ip from the output and create a list as a text file with one ip per line as top50.txt
- Generate a SiLK set file from the text file:

```
rwsetbuild top50.txt top50.set
```

- Use this set to prefilter the flow records to be fed into different tools.
  - rwfilter --anyset top50.set honeypot.rw --pass=stdout | rwcut

Determine the Country Codes for the Top50

```
# rwstats --fields=sip,scc --top --count 51 /data/honeypot.rw
INPUT: 38240 Records for 2422 Bins and 38240 Total Records
OUTPUT: Top 51 Bins by Records
                      sIP|scc|   Records|  %Records|   cumul_%|
            137.184.35.111| us|     17317| 45.285042| 45.285042|
              92.118.39.92| ro|      1701|  4.448222| 49.733264|
              92.118.39.90| ro|      1185|  3.098849| 52.832113|
            185.225.74.213| de|       800|  2.092050| 54.924163|
              92.118.39.91| ro|       629|  1.644874| 56.569038|
…
```

Can I label the sIP with something other than Country Codes?

Yes, Yes you can.

SiLK uses dictionaries called pmaps that allow name value pairs to be defined and understands how IP ranges work.

I have included a pmap that provides that summary of AS to IP associations from the BGP routing table from a date where the honeypot was collecting data. The file is called: 20230401.bgp.pmap.

To use the pmap with rwstats you have to reference it and identify what you are making the association with.

Capturing the ASN associated with the IP

```
# rwstats --pmap-file=asn:20230401.bgp.pmap --fields=sip,src-asn \
--top --count 5 /data/honeypot.rw
INPUT: 38240 Records for 2422 Bins and 38240 Total Records
OUTPUT: Top 5 Bins by Records
              sIP|   src-asn|   Records|  %Records|   cumul_%|
    137.184.35.111|   AS14061|     17317| 45.285042| 45.285042|
      92.118.39.92|   AS47890|      1701|  4.448222| 49.733264|
      92.118.39.90|   AS47890|      1185|  3.098849| 52.832113|
    185.225.74.213| AS3.14644|       800|  2.092050| 54.924163|
      92.118.39.91|   AS47890|       629|  1.644874| 56.569038|
```

In this command we are associating the pmap information with a field labeled asn, we seek to identify the source asn to correspond with the source IP address, so we use the "src-" prefix.

Can we do time series of the data?

```
# rwcount --bin-size=86400 honeypot.rw
               Date|        Records|              Bytes|          Packets|
2023/01/24T00:00:00|         697.30|          2169465.17|          8155.08|
2023/01/25T00:00:00|        2264.08|          5354385.27|         18491.68|
2023/01/26T00:00:00|        3198.40|          6592853.66|         30928.20|
2023/01/27T00:00:00|        4415.23|          9152329.90|         43175.04|
2023/01/28T00:00:00|        2919.00|          5461649.00|         24424.00|
2023/01/29T00:00:00|        3556.00|          6643427.00|         32711.00|
2023/01/30T00:00:00|        5887.00|         14162301.00|         91112.00|
2023/01/31T00:00:00|        5563.55|         14114821.30|         87754.49|
2023/02/01T00:00:00|        3760.45|          8882637.70|         59736.51|
2023/02/02T00:00:00|        3241.00|          7755126.00|         56551.00|
2023/02/03T00:00:00|        2738.00|          3822399.00|         25377.00|
```

*bin size is measured in seconds.

What about a time series of the Top50?

```
# rwfilter --anyset top50.set honeypot.rw --pass=stdout | rwcount --bin-size=86400
                Date|        Records|               Bytes|          Packets|
2023/01/24T00:00:00|         405.30|          1846168.17|          5741.08|
2023/01/25T00:00:00|        1328.70|          4290164.83|         11793.92|
2023/01/26T00:00:00|        1502.00|          3820269.00|         10938.00|
2023/01/27T00:00:00|        2347.00|          5424285.00|         18707.00|
2023/01/28T00:00:00|        1320.00|          2963044.00|          7599.00|
2023/01/29T00:00:00|        1322.00|          2945821.00|          8467.00|
2023/01/30T00:00:00|        3685.00|         10695918.00|         68171.00|
2023/01/31T00:00:00|        2749.55|          9013165.30|         53878.49|
2023/02/01T00:00:00|        2268.45|          6188380.70|         41614.51|
2023/02/02T00:00:00|        1696.00|          4602130.00|         34566.00|
2023/02/03T00:00:00|        1269.00|          1189859.00|          8095.00|
```

Looking at these results we see that there are days with significantly different values than the complete capture. By using a bin size of 300 we see that traffic starts later than the previous result and there are times with no data that had data before.

What about a time series of a specific AS?

```
# rwfilter --pmap-file=asn:20230401.bgp.pmap --pmap-any-asn=AS47890\ honeypot.rw --pass=stdout | rwcount
--bin-size=300
               Date|        Records|               Bytes|          Packets|
2023/01/24T16:10:00|          23.75|            84656.51|           134.25|
2023/01/24T16:15:00|           0.25|             1309.49|             6.75|
2023/01/24T16:20:00|           0.00|                0.00|             0.00|
2023/01/24T16:25:00|           0.00|                0.00|             0.00|
2023/01/24T16:30:00|           0.00|                0.00|             0.00|
2023/01/24T16:35:00|           0.00|                0.00|             0.00|
2023/01/24T16:40:00|           0.00|                0.00|             0.00|
2023/01/24T16:45:00|           0.00|                0.00|             0.00|
2023/01/24T16:50:00|           0.00|                0.00|             0.00|
2023/01/24T16:55:00|           0.00|                0.00|             0.00|
2023/01/24T17:00:00|           0.00|                0.00|             0.00|
```

…

In this command we are filtering on a specific AS value that can be either the source or destination and then piping the output to rwcount. This shows that the hosts from AS47890 didn't start accessing the honeypot until 16:10 on 1/24 and finished before 16:20.